# HCC1395: somatic mutation and RNA evidence

Whole source-call audit | matched-normal DNA baseline | original tumor RNA | pinned reference ORFs. Counts below describe the specified public call sets and RNA run, not every biological mutation or a validated full-length isoform.

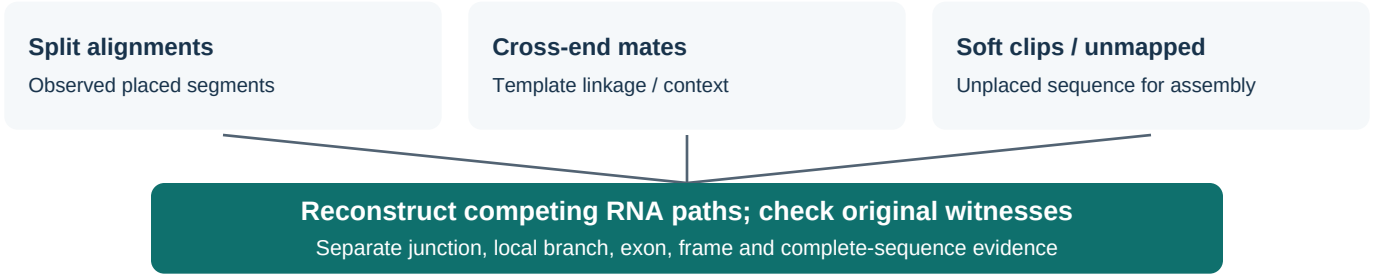
## Small variants: DNA to RNA and local protein




*Boxes show subset order; widths are schematic, not scaled to count.*

## Structural variants: complementary routes to RNA reconstruction

1,788 nominated SV records; 555 have RNA near an end




The legacy split/mate screen is one route, not an eligibility gate.

Soft-clip counts were not enumerated in the old atlas; this channel is not zero.

Breakend-window coverage is locus coverage. Split reads, cross-end mates and soft clips are complementary discovery evidence; none alone resolves a coding frame. Exon overlap is an annotation, not an eligibility filter. The older exon-restricted split/mate screen and selected junction replay are documented later without treating their negatives as absence. The newer COLO829 workflow uses all retained paths and exon-anchored clips.

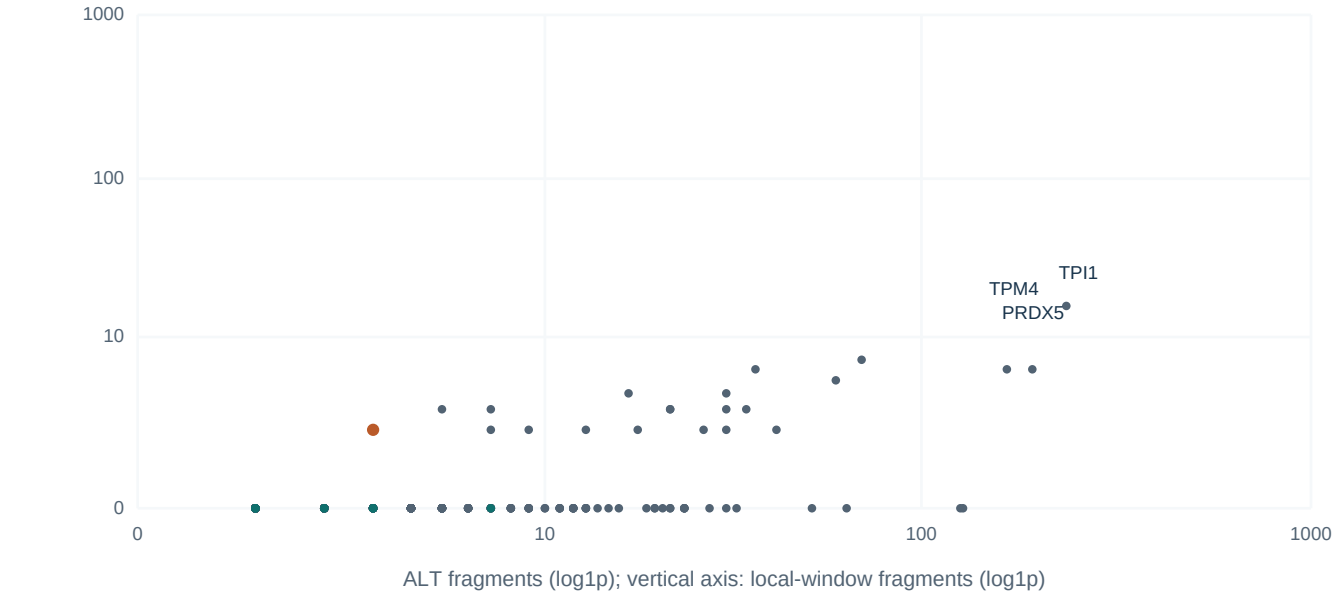
## Variant landscape: depth, coding recovery and unresolved categories

Every nominated small variant is counted above. The scatter shows ALT-observed calls; points on the baseline have no local protein context under the recorded settings. A context may be reference-identical. Separate cultures and assays are never pooled.

### DNA inventory composition and original-RNA outcomes

| | DNA rows | ALT calls | Local context |
|---|---|---|---|
| SNV | 39,447 | 333 | 21 |
| insertion | 157 | 1 | 1 |
| deletion | 1,468 | 9 | 0 |

### RNA allele support versus support for the top local protein window




| Category | What this report can establish |
|---|---|
| Small variants | Original-RNA allele observations and bounded local coding windows |
| SVs / fusions | Original joins and competing RNA paths; annotated coding frame only where resolved |
| ncORFs | Reference coding-block overlaps and RNA ALT; mutant ncORF proteins remain unresolved |
| Viral / other antigen sources | Not surveyed in these host-focused inputs; no absence claim |

## At a glance

| Evidence layer | Count | Interpretation |
|---|---|---|
| Ingested small variants | 41,072 | 39,447 SNVs; 1,625 indels |
| Ingested SV source records | 1,788 | DNA breakpoint records/events, counted in their source sets |
| SVs involving an annotated exon | 354 | Span for DEL/DUP/INV; nominated breakpoint or insertion flanks otherwise |
| Any annotated exon involved | 1,728 | At least one changed base or insertion flank overlaps an Ensembl 87 exon |
| Ensembl 87 coding effect | 491 | Any annotated protein-coding transcript, reference model |
| orfdata ncORF coding-block overlap | 17 | Candidate reference ORF overlap; distinct from mutant translation |
| Original RNA ALT-observed calls | 343 | 2,681 ALT reads; 2,681 ALT fragments |
| Calls with local protein contexts | 22 | 119 ranked contexts; full-length isoforms unresolved |

## Coding annotation and its denominator

Of 41,072 DNA small-variant records, 491 change the coding sequence of at least one Ensembl 87 transcript. The pinned orfdata 0.34.1 GRCh38 catalog (7,264 reference ncORF models; release 2026-10-11.1) has exact coding-block overlaps at 17 calls. The groups overlap at 3 calls, so the union is 505. These are annotation opportunities. The catalog does not assess a cell-line-specific mutant ORF, RNA expression, or peptide evidence.

| Reference annotation | All DNA calls | With original RNA ALT |
|---|---|---|
| Ensembl coding effect | 491 | 135 |
| orfdata coding-block overlap | 17 | 5 |
| Union of either category | 505 | see per-call table |

SNVs query their changed reference base. Deletions query removed reference bases; insertions use the VCF anchor as a candidate overlap. The per-call table preserves that rule. A coding-block hit does not prove the inserted sequence is in frame.

Any annotated exon is involved in 1,728 calls (4.2% of the small-variant source set). This includes UTR and noncoding-transcript exons. Deletions query every removed reference base, including exon crossings with intronic endpoints; insertions query both neighboring reference bases as candidate exon involvement.

## Original RNA and local protein evidence

RNA assay: SRR23456709, ONT direct RNA. Each nominated small variant has a status from this run. An ALT read is a sequence observation; it is not a full-length mutant isoform.

| RNA allele status | Small-variant records |
|---|---|
| alternate observed | 343 |
| reference only | 239 |
| reference and other only | 53 |
| other allele only | 19 |
| alignment without allele call | 7,300 |
| no alignment overlap | 33,118 |

The 343 ALT-observed calls contribute 2,681 allele-read and 2,681 fragment observations. Isovar resolves at least one read-compatible **local** protein context for 22 calls (119 ranked rows). The remaining 321 ALT-observed calls have no local translation under the stated method. No-overlap and uncallable statuses cannot establish absence of RNA expression.

OSTC F9L and PRDX5 F157L have original-read local contexts. An independent proteomic study observed the mutant residues, not these entire contexts. The bounded ONT cDNA fixture is excluded from whole-run counts.

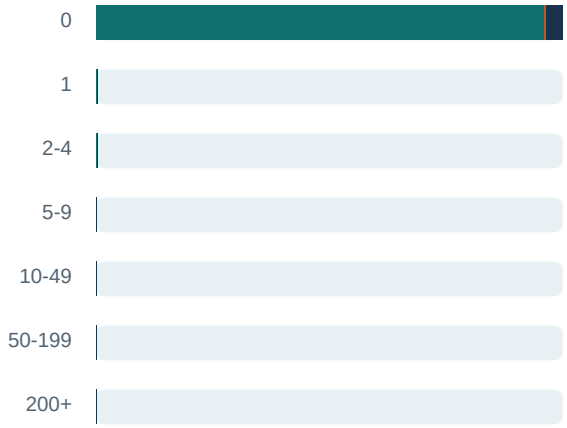
## Structural-variant RNA screen

| Breakend screen outcome | Source records |
|---|---|
| No breakend-window alignment | 1,233 |
| One end covered; no split candidate | 149 |
| Both ends covered; no split candidate | 400 |
| Split candidate; path unresolved | 6 |

Six Talsania records have same-read split-alignment candidates (17 record-level observations). Published POU2F1-NME7 has one candidate read. Junction bases, orientation, splice choice and ORF remain unresolved; no fusion protein is called.

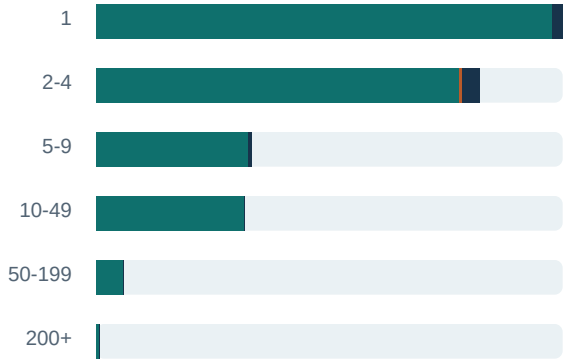
# RNA alternate-read support by variant class

## ALT reads per variant: all records




| | SNV | insertion | deletion |
|---|---|---|---|
| 0 | 39,114 | 156 | 1,459 |
| 1 | 132 | 0 | 3 |
| 2-4 | 105 | 1 | 5 |
| 5-9 | 44 | 0 | 1 |
| 10-49 | 43 | 0 | 0 |
| 50-199 | 8 | 0 | 0 |
| 200+ | 1 | 0 | 0 |

## ALT reads per variant: positive support only

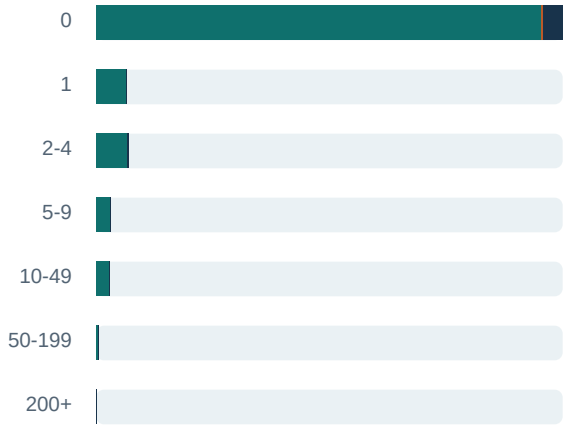



| | SNV | insertion | deletion |
|---|---|---|---|
| 1 | 132 | 0 | 3 |
| 2-4 | 105 | 1 | 5 |
| 5-9 | 44 | 0 | 1 |
| 10-49 | 43 | 0 | 0 |
| 50-199 | 8 | 0 | 0 |
| 200+ | 1 | 0 | 0 |

The positive-support panel excludes zero-ALT records. Bars stack SNV, insertion, deletion and other-replacement record counts. Matching colored columns print each class count directly. These are variant counts, not read counts.

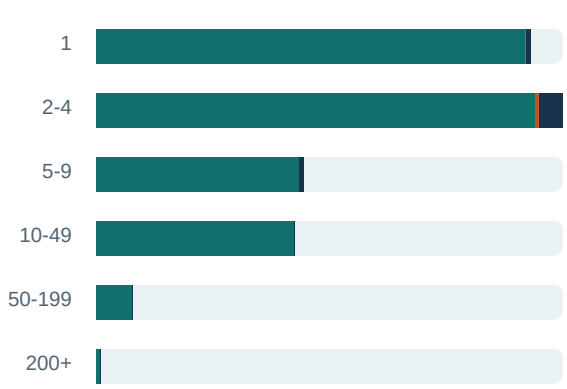
# RNA ALT reads: exon-involved small variants

## ALT reads per exon-involved variant: all records




| | SNV | insertion | deletion |
|---|---|---|---|
| 0 | 1,371 | 4 | 61 |
| 1 | 93 | 0 | 1 |
| 2-4 | 95 | 1 | 5 |
| 5-9 | 44 | 0 | 1 |
| 10-49 | 43 | 0 | 0 |
| 50-199 | 8 | 0 | 0 |
| 200+ | 1 | 0 | 0 |

## ALT reads per exon-involved variant: positive support only




| | SNV | insertion | deletion |
|---|---|---|---|
| 1 | 93 | 0 | 1 |
| 2-4 | 95 | 1 | 5 |
| 5-9 | 44 | 0 | 1 |
| 10-49 | 43 | 0 | 0 |
| 50-199 | 8 | 0 | 0 |
| 200+ | 1 | 0 | 0 |

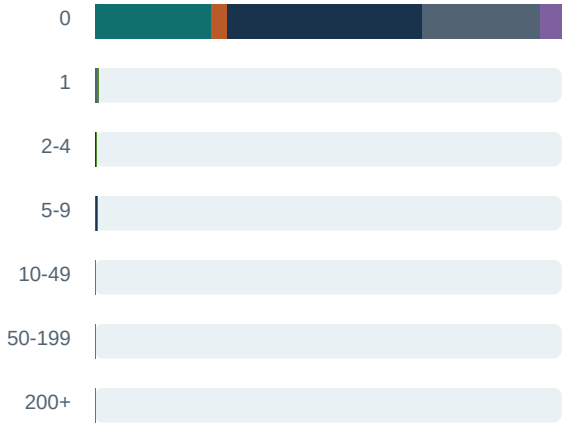
The input is the exon-involved subset of the small-variant source calls. The first panel retains zero-ALT calls; the second conditions on positive ALT reads. Insertion flanks are candidate exon involvement. SVs have a separate plot because a split alignment is not an ALT-spanning read.

## Exon-involved nominated SVs: RNA breakpoint screen

354 of 1,788 nominated SV source records involve an Ensembl 87 exon. Intrachromosomal DEL/DUP/INV records query their full nominated span; INS queries insertion flanks; TRA/BND query nominated breakpoints. 0 records have insufficient breakpoint coordinates to rule out exon involvement and are excluded from this plot.

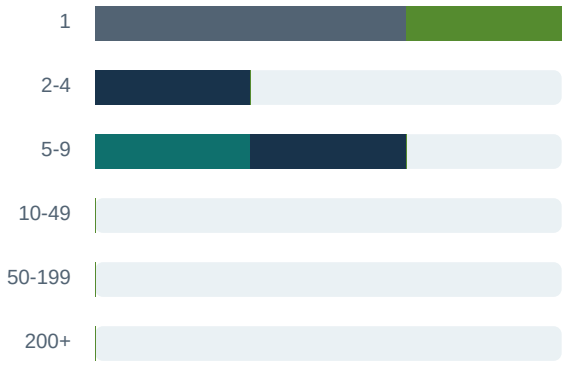
| RNA screen | Exonic SV records | Candidate reads or pairs |
|---|---|---|
| Same-read split alignments | 6 | 17 |
| Cross-end primary mates | 0 | 0 |

### RNA link candidates per exon-involved SV: all records



| | DEL | INS | DUP | INV | TRA | BND |
|---|---|---|---|---|---|---|
| 0 | 87 | 12 | 145 | 88 | 16 | 0 |
| 1 | 0 | 0 | 0 | 2 | 0 | 1 |
| 2-4 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5-9 | 1 | 0 | 1 | 0 | 0 | 0 |
| 10-49 | 0 | 0 | 0 | 0 | 0 | 0 |
| 50-199 | 0 | 0 | 0 | 0 | 0 | 0 |
| 200+ | 0 | 0 | 0 | 0 | 0 | 0 |

### RNA link candidates per exon-involved SV: positive support only



| | DEL | INS | DUP | INV | TRA | BND |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 2 | 0 | 1 |
| 2-4 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5-9 | 1 | 0 | 1 | 0 | 0 | 0 |
| 10-49 | 0 | 0 | 0 | 0 | 0 | 0 |
| 50-199 | 0 | 0 | 0 | 0 | 0 | 0 |
| 200+ | 0 | 0 | 0 | 0 | 0 | 0 |

Bars count SV source records by reported class, binned on split-read plus cross-end mate candidate counts. Same-read supplementary alignments and opposite primary mates are different kinds of screening evidence; neither validates a junction sequence or translated product. The screen uses 500-base windows and mapping quality at least 20. The COLO829 benchmark call sets remain separate in the underlying table.

## Original-read replay: junction versus RNA path

This selected-event replay uses oriented DNA adjacencies and actual RNA alignment records, including soft-clipped bases and low-MAPQ pieces. Direct junction fragments count reads crossing the same RNA join, even when their wider transcript sequences differ. MAPQ 20 is a sensitivity subset, not a gate on whether the junction sequence was observed. Isovar extends these joins through overlapping original read sequences; one read need not span the assembled context. Context length is a bounded assembly result, not an isoform-support count.

| Nominated DNA record | Class | RNA join, all MAPQ | Join, MAPQ 20 subset | Longest context (bp) |
|---|---|---|---|---|
| chr13:52,688,329-60,966,095<br>**SUGT1**<br>chr13:52688329-52688329<br>chr13:60966095-60966095 | DEL | 4 | 3 | 2430 |
| chr21:42,975,947-43,035,721<br>**PKNOX1**<br>chr21:42975947-42975947<br>chr21:43035721-43035721 | DUP | 2 | 2 | 2225 |
| chr9:83,969,153-84,015,113<br>**HNRNPK**<br>chr9:83969153-83969153<br>chr9:84015113-84015113 | DUP | 7 | 7 | 2198 |

Gene names overlap the source breakends (Ensembl 87); they do not establish a coding fusion. IGV loci are one-based inclusive, GRCh38, with no digit separators.

The chr9 duplication has seven direct junction fragments; three event paths survive assembly and the search reaches its ten-path cap. The chr13 deletion has four direct fragments, including one lost at MAPQ 20; chr21 has two. Overlapping reads extend each candidate context; their local branch support still needs evaluation. The three chr1 screen rows reuse one complex read and were not oriented here.

The HCC1395 replay BAM is a source-derived 309-record regional fixture; COLO829 uses the 129,534-record selected-site BAM. Both were extracted after aligning the complete original RNA run. No transcript models were supplied in this geometry audit, so frame and SV-derived protein status remain unresolved. Other splicing and DNA-to-RNA causation have not been excluded.

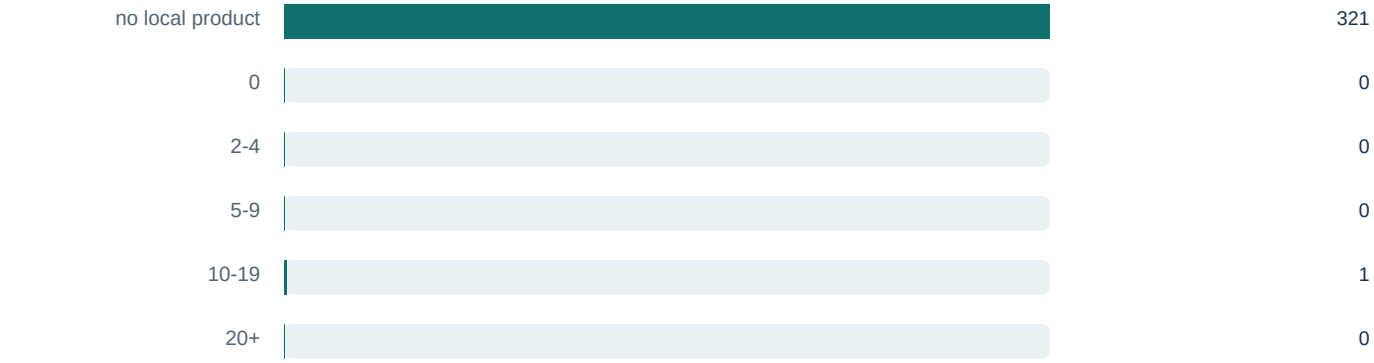
## Isovar-marked amino-acid span

For each RNA ALT-observed call, count amino-acid positions in the mutation interval of its top-ranked local protein context. This is not an independently verified count of changed residues: synonymous DNA calls can produce a reference-identical amino acid window. Calls with no local context are shown explicitly. The second panel omits the one-residue category to expose the longer tail; its denominator is smaller. A zero-length interval can represent a deletion junction. These windows do not determine a complete frameshift tail or a full-length isoform.

### Marked AA positions: all ALT-observed calls




### Marked AA positions: excluding one-AA calls



## Highest-support SNVs with local coding context

Top 3 of 21 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

### TPI1 | chr12:6867653:G>C

IGV (GRCh38): chr12:6867653-6867653 | Silent p.L66= | 236 ALT fragments

16 reported window fragments; 20 aa. Transcript(s): ENST00000229270;ENST00000396705;ENST00000613953

| | |
|---|---|
| REF | KQSLGELIGTLNAAKVPADT |
| RNA | KQNLGSSSALSTRPRCRPTP |

RNA differences have unclassified provenance; DNA-codon-anchored comparison only. Reference: ENST00000229270, residue 56. No unobserved sequence is filled from reference.

### PRDX5 | COSM32759

IGV (GRCh38): chr11:64320895-64320895 | Substitution p.F157L | 193 ALT fragments

6 reported window fragments; 20 aa. Transcript(s): ENST00000265462

| | |
|---|---|
| REF | VRLLADPTGAFGKETDLLLD |
| RNA | VRLLADPTGALGKETDLLLD |

Differing focal RNA residues highlighted. Reference: ENST00000265462, residue 147. No unobserved sequence is filled from reference.

### TPM4 | COSM32783

IGV (GRCh38): chr19:16093699-16093699 | Substitution p.E240Q | 166 ALT fragments

6 reported window fragments; 20 aa. Transcript(s): ENST00000300933;ENST00000344824

| | |
|---|---|
| REF | DKLKEAETRAEFAERTVAKL |
| RNA | DKLKEAETRAQFAERTVAKL |

Differing focal RNA residues highlighted. Reference: ENST00000300933, residue 194. No unobserved sequence is filled from reference.

## Top RNA-supported frameshift nominations

Top 1 of 1 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

### C8orf76 | chr8:123220221:A>AGGGCTAC

IGV (GRCh38): chr8:123220221-123220221 | FrameShift p.L342fs | 3 ALT fragments

2 reported window fragments; 25 aa. Transcript(s): ENST00000276704

| | |
|---|---|
| **REF** | **HPEVKCVGSVALTALVTVSSEEFED** |
| **RNA** | **HPEVKCVGSVARSPDCLGDCILRRI** |

Differing focal RNA residues highlighted. Reference: ENST00000276704, residue 331. No unobserved sequence is filled from reference.

## Top other insertions, deletions and replacements

Top 3 of 9 ALT-supported calls in this category, ranked by source-library ALT fragments. Window support is reported separately. A reference-identical sequence is a coding context, not a neoORF. DNA effect labels and the displayed reference may use different transcript isoforms.

### SFXN1 | chr5:175528598:GA>G

IGV (GRCh38): chr5:175528598-175528599 | ThreePrimeUTR 3' UTR | 7 ALT fragments

RNA ALT observed, but no local protein context was reconstructed. No protein sequence is inferred.

### PPIG | chr2:169641207:ATTATATTTTTTTAATGAAGTTTAGAAAAAAAGCTG>A

IGV (GRCh38): chr2:169641207-169641242 | ThreePrimeUTR 3' UTR | 3 ALT fragments

RNA ALT observed, but no local protein context was reconstructed. No protein sequence is inferred.

### CNOT9 | chr2:218594451:GTCTTGAGGACCTGGGCTCCCTCTGCTA>G

IGV (GRCh38): chr2:218594451-218594478 | ThreePrimeUTR 3' UTR | 3 ALT fragments

RNA ALT observed, but no local protein context was reconstructed. No protein sequence is inferred.

## SV-derived RNA transcripts: junctions, branches and unresolved frames

These are sequence-supported RNA paths. A breakpoint-associated transcript is not automatically a coding fusion or a DNA-causal product. Split placement, soft clips, original query sequence and exon anchors are complementary evidence.

### HNRNPK | chr9_83969153_chr9_84015113_-+

IGV (GRCh38): chr9:83969153-83969153 chr9:84015113-84015113

7 original junction fragments; 3 pinned example RNA paths; representative local-window support 3 fragments. These measures are not pooled.

RNA junction context (nucleotides; no protein frame assigned):

`GGGCGTGCAAAACTAACATTGAAACAGTGAAAAAGAGGATTGCT`
The pinned exon-path audit retains ordinary-transcript and ambiguous-frame interpretations. No event-linked protein is established by these paths.

### SUGT1 | chr13_52688329_chr13_60966095_+-

IGV (GRCh38): chr13:52688329-52688329 chr13:60966095-60966095

4 original junction fragments; 1 pinned example RNA paths; representative local-window support 1 fragments. These measures are not pooled.

RNA junction context (nucleotides; no protein frame assigned):

`CTTCTTCTTTAAGTCTTTTATCTTCCCAGAGTATTAAAGCATTC`
The pinned exon-path audit retains ordinary-transcript and ambiguous-frame interpretations. No event-linked protein is established by these paths.

### PKNOX1 | chr21_42975947_chr21_43035721_-+

IGV (GRCh38): chr21:42975947-42975947 chr21:43035721-43035721

2 original junction fragments; 2 pinned example RNA paths; representative local-window support 1 fragments. These measures are not pooled.

RNA junction context (nucleotides; no protein frame assigned):

`TGGTCTCTACTAAAAATAAAAAATGTGAACCTTGCCAGTATAG`
The pinned exon-path audit retains ordinary-transcript and ambiguous-frame interpretations. No event-linked protein is established by these paths.

## Coding-fusion exemplar

**EIF3K-CYP39A1:** separate supplied-fusion input resolves a 531-aa coding candidate with 149 Illumina junction fragments and independent ONT support. The appended deep dive shows full parental reference alignments and exact per-codon/assembly coverage. It is outside the benchmark-SV funnel.

## ncORFs and other antigen-source categories

Reference ncORF overlap nominates an alternative coding model; it does not establish that the ORF is translated, mutated in-frame, or covered over its complete coding sequence. No mutant ncORF protein was reconstructed here.

| Top overlap / IGV (GRCh38) | ALT fragments | Reference ORF class | Protein status |
|---|---|---|---|
| **NELFB**<br>chr9:137255474:G>C<br>chr9:137255474-137255474 | 22 | intORF | Unresolved mutant ncORF |
| **UQCC1**<br>COSM32805<br>chr20:35384132-35384132 | 5 | intORF | Unresolved mutant ncORF |
| **AC106782.20, BCL7C, MIR762HG**<br>chr16:30894580:G>C<br>chr16:30894580-30894580 | 4 | lncRNA | Unresolved mutant ncORF |

5 ncORF-overlap calls have ALT RNA; the table shows up to 3. Full ORF model identifiers remain in reference_coding_orfdata_annotations.tsv.gz. A canonical protein context at the same locus does not validate the ncORF.

## Assessment coverage - unknown is not absent

| Category | State | What is missing |
|---|---|---|
| Viral RNA / viral ORFs | Not assessed | No viral-aligned RNA/reference panel in this report |
| General cryptic initiation / retained introns | Not comprehensively assessed | Transcript discovery and initiation evidence beyond selected paths |
| Targeted cancer/testis antigen isoforms | Not assessed | Source-qualified isoform coverage audit |
| Germline / other somatic phasing | Not classified beyond available paths | Matched genotype and explicit linkage are required for colored attribution |
| Translated protein / HLA presentation | Not measured by these RNA data | Independent proteomic / immunopeptidomic evidence |

## Sources, processing and limits

All RNA libraries, DNA nominations, polished inputs and references across the report set are listed in SEQUENCING_SOURCES.md. Focused CDKN2A and EIF3K-CYP39A1 reports show highlighted reference alignments and original-RNA support separately from these atlas summaries.

| Input | Pinned scope |
|---|---|
| Matched-normal small variants | SEQC2 v1.2.1 SNV + indel, GRCh38.d1.vd1; HCC1395BL normal |
| SV benchmark | Talsania et al. 2022 Table S4, HCC1395 vs HCC1395BL |
| Original RNA | PRJNA934933 / SRR23456709; 1,049,452 ONT direct RNA reads |
| Alignment | minimap2 2.26 -ax splice -uf -k14; NCBI GRCh38 no-alt |

## Reference model and interpretation

Small-variant coding effects use Varcode with Ensembl release 87 on GRCh38. The human orfdata 0.34.1 catalog uses Ensembl 101 GRCh38 reference ORFs (7,264 models, dataset 2026-10-11.1). An orfdata coding-block overlap is a reference annotation candidate; this analysis did not translate mutated ncORFs from the RNA reads. Genomic transcript overlap is not RNA-compatible isoform support. Original-read allele counts and local protein contexts are separate observations.

## SHA-256 input pins

**SEQC2 SNV VCF gzip**

`0c49f7e1519fc7860b773f79bdcf07453fe4def8303707f34542471d92494c9d`
**SEQC2 indel VCF gzip**

`db7dc8be2ec04bab830ff1411146c7d538c217ff684e874b85d98a3dbbda91cf`
**Talsania Table S4 xlsx**

`11e8d6393b83f0e0b7fdb0b13a57376663fc9662116d78c89a0d5fa530df096b`
**SRR23456709 SRA archive**

`d69db8e48bbb20665c3e03a1fa35742be818a0127994e015cb3a560afae39ac8`
**GRCh38 no-alt FASTA (uncompressed)**

`9cce8b926416dd96b152deea85188495b75f7ac8d634cc723a017067be8702b7`
**orfdata bundled GRCh38 catalog**

`06bc0bf3f81c25a7356f71a38581ff5704c05dcb75752454bc1ca75075ab8c09`
**Ensembl 87 GRCh38 protein FASTA gzip (self screen)**

`d873c199d5dbf20869db3f2cbd6ac726259215d91c2782f9f1ddcad4bb117cf9`

## What remains unresolved

ONT raw-read errors can create other-base calls and competing local contexts. The RNA and DNA libraries come from separate studies. A 500-base breakend window and split-alignment candidate are screening evidence only. Selected junctions were resolved with original reads, but complete isoform assignment and SV protein reconstruction remain unresolved.

Machine-readable per-call annotation, allele, protein, and SV tables and complete reconstruction commands are published with the Isovar HCC1395/COLO829 reports. These PDFs summarize selected public source call sets, not a complete catalogue of every cell-line mutation.

# HCC1395 | EIF3K-CYP39A1 fusion

531-aa complete coding candidate - RNA reconstruction, not measured translation

## 149 exact Illumina junction fragments

### 9 ONT coding-span reads; 4 also have the exact junction

The reference-guided 1,596-nt coding RNA path is tiled by exact original reads, including 19 mapped Illumina reads with overlaps of at least 20 nt. The Illumina input is fusion-enriched. Counts do not measure whole-library prevalence.

**IGV (GRCh37): chr19:39123318-39123318 chr6:46607405-46607405**

### Protein architecture and reference alignment

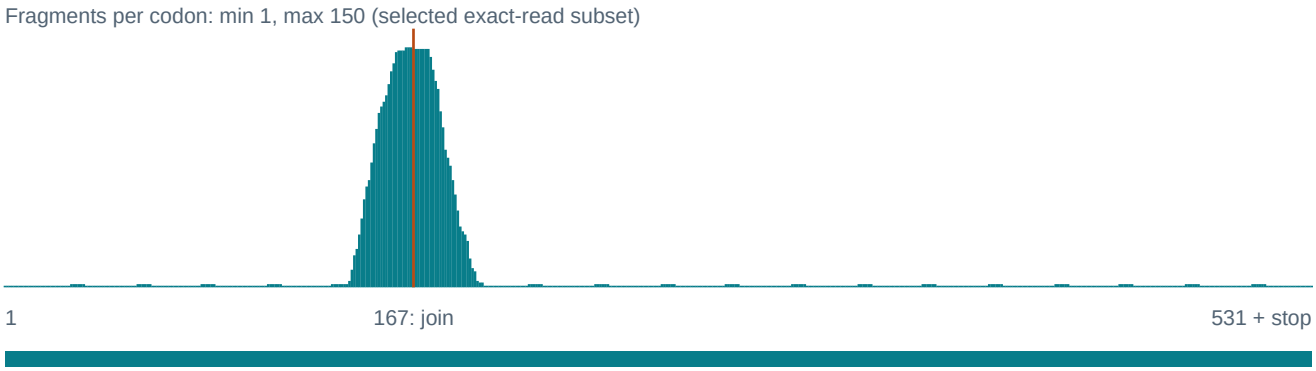
Fusion residues 1-166 match EIF3K residues 1-166. Residue 167 is the split junction codon: one EIF3K base plus two CYP39A1 bases. Its alanine matches CYP39A1 residue 105. Fusion residues 168-531 match CYP39A1 residues 106-469, followed by a stop.

**Junction zoom: aligned by the pinned exon/CDS map**

| | |
|---|---|
| EIF3K 151-186 | QHIDRWLLAEMLGDLSDSQLKVWMSKYGWSADESGQ |
| Fusion 151-186 | QHIDRWLLAEMLGDLSASIPKNVFLALHEKLYIMLK |
| CYP39A1 89-124 | KKVDFELAVQNIVYRTASIPKNVFLALHEKLYIMLK |

The mutant highlight marks junction codon 167. The novel feature is the joined protein context; the CYP39A1-derived suffix is reference sequence, not a novel frameshift tail. Other reported fusion isoforms remain possible.

### Exact original RNA support across the entire coding path




Assembly track: all 1,596 coding bases RNA-witnessed; 0 reference-filled bases.

Each bar counts distinct original fragment IDs with an exact read covering the whole codon. It is not raw BAM depth. The RNA library and ONT library remain separate. Six of 21 ONT candidate reads contain the exact 20+20-base junction; nine span coding start to stop in local collinear alignments. Four satisfy both checks.

# HCC1395 | EIF3K-CYP39A1 fusion

Complete protein aligned to its two reference contributors

D = retained EIF3K reference; F = fusion candidate; A = retained CYP39A1 reference. Dashes mean the contributor is absent at that fusion position, not a deletion from the other gene. The fusion row distinguishes the CYP39A1 suffix and split junction codon. Coordinates at left refer to the fusion protein. The terminal * is the observed stop.

Fusion amino acids 1-60

```
D  MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
F  MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF
A  ------------------------------------------------------------
```

Fusion amino acids 61-120

```
D  FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
F  FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA
A  ------------------------------------------------------------
```

Fusion amino acids 121-180

```
D  LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLS--------------
F  LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLSASIPKNVFLALHEK
A  ---------------------------------------------ASIPKNVFLALHEK
```

Fusion amino acids 181-240

```
D  ------------------------------------------------------------
F  LYIMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKS
A  LYIMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKS
```

Fusion amino acids 241-300

```
D  ------------------------------------------------------------
F  LFSTNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKS
A  LFSTNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKS
```

Fusion amino acids 301-360

```
D  ------------------------------------------------------------
F  AKDNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKA
A  AKDNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKA
```

Fusion amino acids 361-420

```
D  ------------------------------------------------------------
F  IMEGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYI
A  IMEGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYI
```

Fusion amino acids 421-480

```
D  ------------------------------------------------------------
F  IPSGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARW
A  IPSGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARW
```

Fusion amino acids 481-531

```
D  -------------------------------------------------
F  FALLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI
A  FALLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI
```

Terminal stop: * | Annotated EIF3K start and CYP39A1 stop retained.

# HCC1395 | EIF3K-CYP39A1 fusion

Full reference proteins and provenance of amino-acid changes

## EIF3K / ENST00000248342

| | |
|---|---|
| 1-60 | MAMFEQMRANVGKLLKGIDRYNPENLATLERYVETQAKENAYDLEANLAVLKLYQFNPAF |
| 61-120 | FQTTVTAQILLKALTNLPHTDFTLCKCMIDQAHQEERPIRQILYLGDLLETCHFQAFWQA |
| 121-180 | LDENMDLLEGITGFEDSVRKFICHVVGITYQHIDRWLLAEMLGDLSDSQLKVWMSKYGWS |
| 181-218 | ADESGQIFICSQEESIKPKNIVEKIDFDSVSSIMASSQ |

## CYP39A1 / ENST00000275016

| | |
|---|---|
| 1-60 | MELISPTVIIILGCLALFLLLQRKNLRRPPCIKGWIPWIGVGFEFGKAPLEFIEKARIKY |
| 61-120 | GPIFTVFAMGNRMTFVTEEEGINVFLKSKKVDFELAVQNIVYRTASIPKNVFLALHEKLY |
| 121-180 | IMLKGKMGTVNLHQFTGQLTEELHEQLENLGTHGTMDLNNLVRHLLYPVTVNMLFNKSLF |
| 181-240 | STNKKKIKEFHQYFQVYDEDFEYGSQLPECLLRNWSKSKKWFLELFEKNIPDIKACKSAK |
| 241-300 | DNSMTLLQATLDIVETETSKENSPNYGLLLLWASLSNAVPVAFWTLAYVLSHPDIHKAIM |
| 301-360 | EGISSVFGKAGKDKIKVSEDDLENLLLIKWCVLETIRLKAPGVITRKVVKPVEILNYIIP |
| 361-420 | SGDLLMLSPFWLHRNPKYFPEPELFKPERWKKANLEKHSFLDCFMAFGSGKFQCPARWFA |
| 421-469 | LLEVQMCIILILYKYDCSLLDPLPKQSYLHLVGVPQPEGQCRIEYKQRI |

Highlighted reference residues contribute to the fusion. Dimmed residues are not retained. The fusion codon at residue 167 contains bases from both partners and encodes CYP39A1 alanine 105; it is not counted as a complete EIF3K-derived codon.

### Variant provenance

- Focal change
- Germline
- Phased somatic

No additional non-reference residues occur in the displayed RNA sequence relative to its indicated reference contributor(s). No germline or other somatic allele is assigned without genotype and phasing evidence. Missing sequence stays unobserved.

There is no reference-filled region in the determined ORF. The gene junction at coding offset 499 is an RNA-supported splice/fusion boundary, not an assembly-to-reference stitching point. Support falls outside the enriched junction region, as shown by the codon track; no junction count is assigned to every residue.

# HCC1395 | EIF3K-CYP39A1 fusion

All sequencing and reference inputs used in this focused report

### Illumina tumor RNA: public INTEGRATE test-data.tar.gz

Original unmapped.bam (185 fusion-enriched records) and accepted_hits.bam (19 mapped records retained for coding tiling). Archive SHA-256: 9a37130e768e424935008f1d0a67050f6e11a242a263ed2f20ba071d94b767cb. The archive is the source identifier; no SRA run accession is asserted for this subset.

### Independent ONT direct RNA: SRR23456709 / PRJNA934933

BioSample SAMN33292477; 1,049,452 reads in the source run. The report checks 21 original candidates and 42 local alignment records. This is a local candidate screen, not a genome-wide uniqueness test.

### Tumor and matched-normal DNA: INTEGRATE example archive

All original alignments in two GRCh37 flanking intronic windows were retained. 18 tumor cross-gene fragments versus 0 normal. This supports a rearrangement, without resolving its exact DNA breakpoint.

### Exon/cDNA and coding references

Pinned GRCh37-lite archive models: EIF3K ENST00000248342 and CYP39A1 ENST00000275016. Ensembl 87 CDS offsets transfer only after exact whole-transcript sequence identity. ONT placements use GRCh38 and separately pinned Ensembl 87 exon coordinates; genomic coordinates from the two assemblies are never mixed.

### Derived files, checksums and broader report source index

The adjacent HCC1395_EIF3K_CYP39A1_fusion.json pins all eight BAM/FASTA/model inputs and contains the complete protein, reference contributors and per-codon support. SEQUENCING_SOURCES.md lists every library, DNA source and reference used in the Sid/HCC1395/COLO829 report set.

The first page highlights strong junction support. It does not assign 149 fragments to every amino acid or claim that 149 reads span the entire protein. Complete coding sequence support is the separately displayed overlap tiling, corroborated by the source-separated ONT checks.